STATEMENT OF WORK (SOW) & MASTER SERVICES RATE CARD

Turnkey Rollout, Site Engineering, Telecom Integration, and 24/7 Managed Operations & Maintenance (O&M)

Author: Oliver Tungol | Version: 2.0 (Expanded Master Services SOW) | Commercial FX Baseline: USD 1.00 = PHP 55.00

EXPLICIT GOVERNANCE: TOWERCO-LED ACQUISITION BASELINE VS. VENDOR CONTINGENCY RATE CARD

1. Primary Operational Baseline (TowerCo-Led Scope): The primary operational objective of the Operator is to utilize Independent Tower Companies (**TowerCos**) under Master Lease Agreements (MLA) — either through **Tier 1 (Existing Co-Location on the 13,746 audited tower database)** or **Tier 2 (TowerCo Build-to-Suit / B2S)**. Under this primary baseline, **all Site Acquisition (SAQ) work** — including candidate hunting, landlord negotiations, title due diligence, lease execution, Barangay clearances, LGU building permits, CAAP clearances, and utility power applications — is **STRICTLY THE SCOPE AND FINANCIAL RESPONSIBILITY OF THE TOWERCO, NOT THE VENDOR**.

2. Candidate Hunting & Mandatory Joint Approval Gate: In day-to-day operations, physical candidate hunting is executed by the **TowerCo**. However, candidate approval is governed strictly by a **Mandatory Joint Approval Gate**: (a) The **Vendor** conducts on-site technical vetting verifying RF line-of-sight (LOS) to the demand settlement centroid, antenna mounting centerline/height, structural loading headroom (EPA), DC power availability, and cable routing feasibility, and issues formal technical endorsement in the TSSR; (b) The **Operator** (TelTech/Comclark) evaluates commercial terms under MLA and issues final approval to proceed. *(No TowerCo candidate shall proceed to deployment without joint Vendor technical endorsement and Operator final sign-off)*.

3. Commercial Purpose of Vendor SAQ Rate Card (Items SAQ-01 to SAQ-08): The unit rates requested under Category 1 are included in the Price Book **STRICTLY AS A STANDBY / CONTINGENCY CALL-OFF RATE CARD ("Operator Self-Acquisition / Own-Build Option")**. The Operator requires firm unit pricing in the event that the Operator elects or is compelled to execute direct self-acquisition and self-build (Tier 3 Operator Greenfield Monopole / Own-Build) — such as in unserved remote search rings where zero TowerCos operate, or where all operational TowerCos issue formal structural denial letters or refuse to construct a Build-to-Suit tower.

CRITICAL BIDDER INSTRUCTION: Bidders shall NOT embed site acquisition, land leasing, or municipal permitting risk markups inside their baseline turnkey hardware, installation (I&C), or co-location pricing. Items SAQ-01 to SAQ-08 are evaluated as a standby rate card and are billable ONLY upon formal written Work Order issued by the Operator for an authorized self-acquired site.

1. PROGRAM OVERVIEW & SITING ARCHITECTURE

The 5G Fixed Wireless Access network utilizes 3GPP Band n50 (1432–1517 MHz, 60 MHz TDD) deployed across high-density suburban and rural barangays. Siting delivery is strictly governed by the **Mandatory 3-Tier Sourcing Hierarchy**:

- **Tier 1 (Mandatory Baseline - 89.7% / 897 Sites):** Co-location on existing operational towers (PTCI/MIDC, EdgePoint, FTAP Globe, FTAP Smart, LDIC) within the designated search ring.
- **Tier 2 (Secondary Target):** TowerCo Build-to-Suit (B2S) where existing structures lack load capacity.
- **Tier 3 (Restricted Exception - 10.3% / 103 Sites):** Direct Operator Greenfield Monopole construction (80x 30m / 23x 40m).

2. DETAILED TECHNICAL SCOPE OF WORK (DEPLOYMENT & ROLLOUT SERVICES)

2.1 Site Acquisition (SAQ) & Legal Permitting Services

The Contractor shall deliver complete turnkey Site Acquisition (SAQ) and permitting services: hunting and ranking three (3) viable candidates per search ring (500m urban / 1,000m rural); compiling the Preliminary Candidate Evaluation Report (PCER); producing the comprehensive Site Acquisition Report (SAR) with WGS84 GPS (6 decimals), cadastral sketches, clean title verification (TCT/OCT), and 360-degree panoramic photos; preparing the TowerCo Space Application (SAA) loading dossier under the MLA; supporting greenfield lease negotiations; and securing all statutory permits: Barangay Clearance, HOA Resolution, LGU Building Permits under PD 1096, CAAP Height Clearance, DENR CNC/ECC, DPWH ROW clearance, CFEI, FSIC, power utility energization (Meralco/Coop), and LGU Certificate of Occupancy.

2.2 Site Engineering & Detailed Technical Site Survey (TSSR)

Rigorous multidisciplinary engineering surveys including: 360-degree drone panoramic photo sweeps at antenna centerlines (30m to 45m AGL); structural audit of steel members, gussets, and anchor bolts; calculation of Effective Projected Area (EPA) and operating weight for 3x 8T8R RRUs, 3x macro antennas, AMBs, and cables; wind stress simulation under TIA-222-H for 250 km/h basic wind speed (Category 5 typhoon); foundation verification; utility transformer kVA and DC power plant headroom audit; battery autonomy calculation (>= 4.0h @ 3.0 kW load); 3-point fall-of-potential ground resistance testing (mandating ≤ 5.0 Ohms); RF line-of-sight and clutter verification (RSRP ≥ -95 dBm); complete AutoCAD Detailed Engineering Design (DED) blueprints signed and sealed by licensed Professional Engineers; and post-construction As-Built CAD documentation.

2.3 Radio Network Planning & Design (RNPD)

Macro-level 5G RF planning utilizing 3D ray-tracing propagation modeling engines calibrated against 5m clutter and DTM data; nominal search ring generation; coverage prediction mapping; capacity dimensioning for Band n50 (60 MHz TDD); and Cell Design Data (CDD) parameter engineering, including cell naming, gNodeB IDs, Physical Cell ID (PCI) planning with modulo-3/modulo-30 separation, Root Sequence Index (RSI), Neighbor Cell Lists (NCL), and eRET profiles.

2.4 Telecom Logistics, Warehousing & Material Handling

End-to-end logistics management: port receiving at Manila/Batangas; bonded climate-controlled storage in Mega Manila and regional hubs (North Luzon, South Luzon, Visayas, Mindanao); serial-level barcode asset tagging in ERP; BOM kitting and cable pre-cutting based on approved TSSR; firmware pre-flashing; line-haul freight; and dedicated final-mile rough-road transport to remote barangay sites utilizing heavy-duty 4x4 trucks, motorized bancas, or manual porterage across unpaved terrain, fully insured.

2.5 Telecom Installation, Testing, Commissioning & Integration (TI)

Rigorous execution by certified riggers and telecom engineers: hoisting and mounting 3x macro high-gain antennas and 3x 8T8R RRUs on tower AMBs; setting mechanical tilts and azimuths (0°, 120°, 240°); installing eRET cables (AISG 2.0); routing armored optical jumpers and DC power cables along cable ladders; installing cold-shrink weatherproof boots; installing Type 2 DC SPDs at tower top and base; bonding 3-point grounding kits to external ground busbar; 19" rack installation of BBU; GPS antenna installation for PTP/SyncE; DC turn-up; software baseline loading; optical transceiver power verification; VSWR sweep testing (VSWR ≤ 1.35:1, Return Loss ≥ 16.5 dB); IP backhaul integration; Core (UPF/AMF) connection; OSS synchronization; alarm clearance; and commercial on-air declaration.

2.6 Single Site Verification (SSV) & Site Acceptance (ATP)

Mandatory three-stage acceptance procedure: ATP-A (Mechanical, Civil & Electrical inspection of bolt torque marks, cable dressing, weatherproofing, grounding resistance ≤ 5.0 Ohms); ATP-B (Active Telecom & Functional verification of 8T8R MU-MIMO operation, remote eRET tilt control, alarm failover simulations); and SSV (stationary throughput testing at near, mid, and cell-edge points per sector validating ≥ 450 Mbps peak DL, ≥ 50 Mbps cell-edge DL, ≥ 15 Mbps cell-edge UL, sector swap checks, handover tests, ping ≤ 18 ms RTT, and signed SSV Report).

2.7 Cluster Radio Network Optimization (RNO)

Multi-site cluster optimization across contiguous clusters of 10–20 sites: pre-launch scanner and test UE drive testing; electrical downtilt (eRET) optimization; azimuth adjustments; PCI re-planning; neighbor list cleanup; handover parameter tuning; and 14-day post-launch OSS statistical validation confirming cluster Radio Availability ≥ 99.95%, Session Setup Success Rate (CSSR) ≥ 99.60%, Session Drop Rate (CDDR) ≤ 0.15%, and cluster average throughput ≥ 450 Mbps.

3. OPERATIONS & MAINTENANCE (O&M) / MANAGED SERVICES SPECIFICATION

The Contractor shall deliver an integrated 24/7/365 Managed Services organization to ensure continuous 99.95% network availability across all 5,000 sites nationwide:

- **24/7/365 Central NOC Surveillance:** Tier-1 real-time surveillance across RAN, Core, Power, and Transmission via EMS/OSS. Automated alarm correlation, deduplication, and trouble ticket creation within 5 minutes. Daily Operational Flash Reports at 06:00 PHT, weekly reviews, and monthly SLA dossiers.
- **Tier-2 / Tier-3 Support:** 24/7 remote diagnostics, soft/hard resets, card re-initialization, config restoration, and transmission failover. Tier-3 OEM TAC Level 3 escalation, bug patch testing, and Root Cause Analysis (RCA) reporting within 24 hours.
- **Field First Line Maintenance (FLM):** Dedicated 24/7 mobile crews in 4x4 vehicles stationed at regional depots. Mandated MTTR: **Severity 1 (Critical) ≤ 2.0h Urban / ≤ 4.0h Rural; Severity 2 (Major) ≤ 4.0h Urban / ≤ 6.0h Rural; Severity 3 (Minor) ≤ 12.0h Urban / ≤ 24.0h Rural.** Includes active hardware swapping, passive electrical triage, and emergency fiber backhaul splicing.
- **Quarterly Preventive Maintenance (PM):** 100% completion p.a. logged via Mobile App: Module A (Telecom Equipment & Optics); Module B (DC Power Plant & LiFePO4 Battery discharge test); Module C (Grounding grid ≤ 5.0 Ohms); Module D (Bi-annual tower climbing & rigging audit); Module E (Compound civil, fencing & weed clearance).
- **Spare Parts Management (SPMS):** Regional depots maintaining an audited 2.0% active hardware float. Guaranteed Advance Hardware Replacement (AHR) Next-Business-Day (NBD) standard or 4-hour emergency delivery. Full RMA reverse logistics lifecycle tracking (TAT ≤ 30 days).
- **Continuous RNO & Interference Hunting:** Ongoing RF KPI surveillance, weekly Worst Performing Cell (WPC) triage, spectrum analyzer sweeps for external jamming/boosters, and PIM testing (≤ -155 dBc).
- **Facility, Power & Software Maintenance:** Tower anti-corrosion painting, plinth crack repair, utility Meralco/Coop coordination, 250-hour generator servicing, maintenance window patch deployments (00:00-05:00), and weekly automated backups.

4. MASTER SERVICES PRICING SCHEDULE & COMMERCIAL RATE CARD

Bidders must quote firm, fixed unit rates in US Dollars (USD) and Philippine Pesos (PHP @ baseline FX 55.00) in the official Excel price book. Quoted unit rates shall serve as the official contract rate card for Phase 1 (1,000 sites), ongoing Managed Services O&M, and all expansion call-offs up to 5,000 sites.

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
CAT–01	**DOMAIN: SITE ACQUISITION & LEGAL PERMITTING**								
SAQ–01	**Nominal Search Ring Siting & Candidate Hunting**	[OPERATIONAL BASELINE: Siting and candidate hunting is executed by the TowerCo. Vendor performs on-site technical vetting (RF LOS, antenna height, structural feasibility) and issues joint endorsement;	Per Search Ring	1,000	5,000	[Bidder Quote]	[Formula]	100% upon Operator formal approval of PCER ranking 3 compliant candidates.	**Approved Preliminary Candidate Evaluation Report (PCER)**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		Operator issues final approval. COMMERCIAL PRICING BASIS: Quoted rate serves strictly as a Standby / Contingency Call-Off Rate Card for Operator Self-Acquisition / Own-Build]. Field identification, screening, and ranking of three (3) viable candidates (Candidate A - Primary, Candidate B - Secondary, Candidate C - Contingency) within the assigned nominal search ring (500m radius for urban/suburban; 1,000m radius for rural). Mandatory spatial screening against audited TowerCo database (13,746 towers) to prioritize Tier 1 co-location. Scope includes visual inspection of antenna mounting feasibility, ground space availability (minimum 2.0m x 2.0m pad), line-of-sight to demand settlement centroid, road access verification (all-weather vs 4x4), and issuance of the Preliminary Candidate Evaluation Report (PCER). • **Inclusions:** Desktop spatial analysis, physical on-ground hunting, 3 candidate identification, GPS logging, PCER compilation. • **Exclusions:** Commercial lease execution (handled by Operator under MLA).							
SAQ–02	**Detailed Site Acquisition Report (SAR) & Technical Dossier**	[OPERATIONAL BASELINE: TowerCo provides primary site attributes and property profile under MLA. Vendor reviews and compiles technical feasibility annex in TSSR; Operator signs off. COMMERCIAL PRICING BASIS: Standby / Contingency Call-Off Rate Card for full standalone SAR compilation during Operator Self-	Per Candidate Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon formal engineering and legal acceptance of the SAR package by Operator.	**Comprehensive Site Acquisition Report (SAR) Dossier**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		Acquisition]. Comprehensive on-ground site acquisition audit and property investigation for the selected primary candidate. Compilation of standardized Site Acquisition Report (SAR) containing: official PSGC code, precision WGS84 GPS coordinates (>= 6 decimal places), ground elevation AMSL, property cadastral boundary sketch, certified land title verification (TCT/OCT/Tax Declaration), lessor profile and contact matrix, preliminary structural condition assessment, preliminary power availability (grid connection point, transformer distance, utility provider), 360-degree panoramic site photos, logistics route access classification, and SAQ risk assessment matrix. • **Inclusions:** On-ground cadastral audit, title verification, lessor interview, 360-deg photo sweep, complete SAR package. • **Exclusions:** Payment of land rental / lease deposits (direct Operator responsibility).							
SAQ–03	**TowerCo Space Application (SAA) & Technical Dossier Support**	[APPLIES TO TOWERCO CO-LOCATION: Preparation of technical equipment loading profile (3x 8T8R RRU, 3x Antennas, hybrid cables, 2m x 2m pad) and coordination with TowerCo engineering until SAA approval. Commercial lease executed directly by Operator under MLA]. Preparation, technical compilation, submission, and engineering follow-through of the TowerCo Space Application (SAA) package for co-location under Operator's Master Lease Agreement	Per Co-Location Site	897	4,517	[Bidder Quote]	[Formula]	50% upon SAA submission; 50% upon TowerCo issuance of formal SAA Approval / NTP.	**Approved TowerCo SAA & Official Notice to Proceed (NTP)**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		(MLA). Compiles exact proposed loading profile: 3x 8T8R RRU, 3x macro antennas/AAUs, antenna mounting brackets (AMB), hybrid trunk cables, DC distribution requirements, and ground plinth reservation (2.0m x 2.0m). Coordinates with TowerCo technical team during structural vetting until formal issuance of SAA Approval and Notice to Proceed (NTP). • **Inclusions:** SAA documentation, loading calculation submission, liaison with TowerCo engineering, tracking SAA to NTP. • **Exclusions:** Commercial MLA negotiations and monthly tower rental fees (exclusive Operator responsibility).							
SAQ–04	**Greenfield Land Lease Negotiation & Notarized Contract Support**	[CONTINGENCY / SELF-BUILD SCOPE: Applicable strictly to Operator Own-Build Greenfield sites where no TowerCo exists or TowerCo denies build. Direct lessor negotiation, title due diligence, and lease notarization]. Direct lessor negotiations for greenfield sites: boundary survey coordination, verification of registered ownership (clean title check at Registry of Deeds, inheritance/estate verification, real property tax clearance), negotiation of long-term lease terms (10-15 years) within Operator budgetary caps, preparation of lease annexes, securing lessor spousal consent and IDs, and facilitation of formal contract notarization and annotation on title. • **Inclusions:** Lessor negotiations, title due diligence, boundary verification, notary liaison, lease execution assistance. • **Exclusions:** Actual	Per Greenfield Site	103	483	[Bidder Quote]	[Formula]	50% upon mutual execution of Lease Agreement; 50% upon Notarization and Title Annotation.	**Executed & Notarized Contract of Lease + Clean Title Dossier**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		lease rentals, advance deposits, and title annotation official fees (Operator pays directly).							
SAQ–05	**Local Community, Barangay & Neighborhood Clearances**	[OPERATIONAL BASELINE: TowerCo secures community/Barangay clearances for TowerCo towers. COMMERCIAL PRICING BASIS: Standby / Contingency Call-Off Rate Card for Operator Self-Acquisition / Greenfield sites]. Securing official local community clearances: presentation to Barangay Council, public hearing facilitation (if mandated), processing and issuance of official Barangay Resolution of Support, Barangay Construction Clearance, and certified Neighborhood / Homeowners Association (HOA) Resolution / Certificate of No Objection from contiguous property owners within a 50-meter radius. • **Inclusions:** Barangay consultations, neighbor signature campaigns, clearance filing and follow-through, official receipts. • **Exclusions:** Official LGU statutory fees (reimbursable at actual cost against official receipts).	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon issuance of certified Barangay Clearance & HOA No Objection Resolution.	**Certified Barangay Clearance & HOA Resolution of No Objection**
SAQ–06	**Local Government Unit (LGU) Building & Construction Permits**	[OPERATIONAL BASELINE: TowerCo secures LGU Building Permits under PD 1096 for TowerCo towers. COMMERCIAL PRICING BASIS: Standby / Contingency Call-Off Rate Card for Operator Self-Build Greenfield Monopoles]. Complete processing and procurement of municipal/city LGU Building Permits in accordance with the	Per Greenfield Site / Co-loc where required	103	483	[Bidder Quote]	[Formula]	100% upon release of official LGU Building Permit and Approved Engineering Plans.	**Official LGU Building Permit & Endorsed Blueprints**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		National Building Code of the Philippines (PD 1096). Scope includes securing Locational Clearance, Zoning Compliance, Architectural Permit, Civil/Structural Permit, Electrical Permit, Sanitary/Plumbing Permit, and Mechanical Permit, backed by complete sets of plans signed and sealed by licensed Professional Engineers (Civil, Structural, Electrical, PEE). • **Inclusions:** Permit application filing, engineering plan liaison, follow-through across city/municipal engineering offices. • **Exclusions:** Official municipal permit assessment fees (reimbursable at face value against official receipts).							
SAQ–07	**Statutory & Environmental Clearances (CAAP, DENR, DPWH)**	[OPERATIONAL BASELINE: TowerCo secures CAAP Height Clearance and DENR CNC/ECC for TowerCo assets. COMMERCIAL PRICING BASIS: Standby / Contingency Call-Off Rate Card for Operator Self-Build Greenfield Monopoles]. Procurement of mandatory national statutory clearances: Civil Aviation Authority of the Philippines (CAAP) Height Clearance Permit / Obstruction Evaluation; Department of Environment and Natural Resources (DENR) Certificate of Non-Coverage (CNC) or Environmental Compliance Certificate (ECC); and Department of Public Works and Highways (DPWH) Road Right-of-Way (ROW) clearance where site access fronts national highways. • **Inclusions:** Technical submittals, coordinate and elevation	Per Site	103	483	[Bidder Quote]	[Formula]	100% upon issuance of official CAAP Height Clearance and DENR CNC/ECC.	**Official CAAP Height Clearance Permit & DENR CNC/ECC**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		certification, CAAP/DENR filing and technical defense. • **Exclusions:** Official statutory assessment fees (reimbursable at actual cost).							
SAQ–08	**Utility Interconnection, CFEI, FSIC & Occupancy Permitting**	[OPERATIONAL BASELINE: TowerCo manages utility service entrance and meter installation on shared compound. COMMERCIAL PRICING BASIS: Standby / Contingency Call-Off Rate Card for Operator Self-Build sites]. Securing post-construction occupancy and power clearances: processing Certificate of Final Electrical Inspection (CFEI), Bureau of Fire Protection (BFP) Fire Safety Inspection Certificate (FSIC), commercial power utility (Meralco / Regional Electric Cooperative) service application, transformer tapping coordination, meter base energization, and municipal LGU Certificate of Final Occupancy / Permit to Operate. • **Inclusions:** BFP inspection facilitation, CFEI filing, utility coordination, meter installation liaison, occupancy release. • **Exclusions:** Utility transformer installation capital contribution and security deposits (paid by Operator).	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	50% upon CFEI/FSIC release; 50% upon commercial grid energization and Occupancy Permit.	**CFEI, FSIC, Utility Meter Energization Notice & Certificate of Occupancy**
CAT–02	**DOMAIN: SITE ENGINEERING & DETAILED SITE SURVEY**								
ENG–01	**Technical Site Survey (TSS) & 360° Drone Panoramic Audit**	Rigorous multi-disciplinary physical site survey executed by licensed telecom and civil engineers: 360° high-resolution panoramic photographic sweep at target antenna centerline (using calibrated drone or mast camera); precision dual-frequency GPS coordinate recording; physical measurement of compound	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon completion of field survey and upload of raw survey package and photos.	**Raw TSS Survey Package & High-Resolution 360° Panoramic Photo Suite**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		dimensions, tower base, shelter/cabinet footprint; cable gantry routing audit; transport access route assessment (bridge load limits, tight turning radii); and compilation of raw TSS data. • **Inclusions:** Multi-disciplinary field team, calibrated drone flight, 360-deg panoramic photos, spatial sketches, access log. • **Exclusions:** Tower climbing above 60m (requires separate high-mast specialized rigging permit if applicable).							
ENG-02	**Structural Loading, Wind Stress & Civil Engineering Assessment**	Engineering structural evaluation of the proposed host structure (tower, rooftop, or monopole): audit of steel member sections, bracing, gusset plates, anchor bolts, and corrosion condition; calculation of total Effective Projected Area (EPA) and weight for new equipment (3x 8T8R RRU, 3x macro antennas/AAUs, AMB brackets, hybrid cable); structural wind load simulation under TIA-222-H for 250 km/h basic wind speed (Category 5 typhoon standard); foundation capacity verification; and formal structural analysis report certified by a licensed Professional Structural Engineer (PE). • **Inclusions:** Structural audit, EPA calculations, TIA-222-H wind simulation, foundation check, PE sign-off. • **Exclusions:** Tower reinforcement civil execution (governed by separate structural modification call-out rate card).	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon submission and acceptance of the PE-Endorsed Structural Analysis Report.	**PE-Signed Structural Engineering Analysis & Loading Report**
ENG-03	**Electrical, DC Power & Earth Grounding Engineering Assessment**	Electrical engineering audit of commercial AC utility power and DC power plant: inspection of transformer capacity (kVA), service	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon Operator acceptance of Electrical & Power Assessment Report.	**Electrical, Power & Grounding Engineering Assessment Report**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		entrance voltage, phase balance, AC main breaker rating; assessment of existing DC power plant (rectifier total capacity, SMPS controller, available DC branch breakers >= 63A for new radio load); calculation of LiFePO4 battery autonomy (minimum 4.0h @ 3.0 kW load, <= 80% DoD); and 3-point fall-of-potential earth resistance measurement of the grounding grid, mandating <= 5.0 Ohms pursuant to Philippine Electrical Code (PEC) and TelTech engineering specifications. • **Inclusions:** Commercial power audit, DC rectifier capacity calculation, battery autonomy simulation, 3-point earth test. • **Exclusions:** Major utility transformer upgrades (handled under utility capex contribution).							
ENG–04	**RF Engineering Siting, Line-of-Sight (LOS) & Clutter Assessment**	On-site RF validation of nominal design coverage over target barangay demand centroid: physical line-of-sight (LOS) inspection at proposed antenna centerlines (30m, 35m, 40m AGL); clutter height and foliage density classification; identification of close-in obstructions within 200m of sector azimuths (0°, 120°, 240°); calculation of first Fresnel zone clearance; and field verification that simulated coverage will achieve target cell-edge RSRP >= -95 dBm and SINR >= 6 dB. • **Inclusions:** On-ground visual LOS audit, clutter verification, azimuth clearance check, RF siting validation report. • **Exclusions:** Continuous cluster drive test (governed under Lot 5 Scope RNO-01).	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon RF Engineering approval of Siting & LOS Report.	**RF Siting & Line-of-Sight (LOS) Clearance Report**

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ENG-05	**Detailed Engineering Design (DED) & Technical Site Survey Report (TSSR)**	Consolidation of multidisciplinary survey data into a complete Detailed Engineering Design (DED) blueprint package and Technical Site Survey Report (TSSR): Architectural Site Layout Plan; Tower Elevation & Antenna Orientation Plan; Antenna Mounting Bracket (AMB) Fabrication Details; Cable Ladder, Gantry & Cable Tray Routing Plan; Electrical Single Line Diagram (SLD); Earth Grounding Ring & Lightning Protection Layout; Site Bill of Quantities (BoQ); and formal endorsement by licensed Professional Engineers (PE-Civil/Structural, PE-Electrical, PECE-Electronics) conforming to NBCP and PEC. • **Inclusions:** AutoCAD DED drawings, multi-PE signed & sealed plans, BoQ compilation, full TSSR dossier submission. • **Exclusions:** Structural redesigns necessitated by Operator-mandated changes to nominal search ring after survey approval.	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon formal joint Operator-TowerCo TSSR approval.	**Approved TSSR Dossier with PE Signed & Sealed CAD Blueprints**
ENG-06	**As-Built Engineering Documentation & Final Completion Dossier**	Comprehensive post-construction field engineering audit capturing all redline installation adjustments: updating AutoCAD drawings to reflect exact as-built antenna heights, azimuths, mechanical tilts, optical cable routes, rack layouts, breaker allocations, and grounding busbar termination points; compiling comprehensive photographic completion book (minimum 50 high-res timestamped photos per site); and securing final Professional Engineer endorsement. • **Inclusions:** Field redline audit, final CAD as-built	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon formal issuance of the Final Acceptance Certificate (FAC).	**Final As-Built CAD Drawings (DWG & PDF) + Photographic Completion Book**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		drawings (DWG & PDF), photographic completion book, PE endorsement. • **Exclusions:** Routine operational documentation updates post-commercial acceptance.							
CAT-03	**DOMAIN: RADIO NETWORK PLANNING & DESIGN**								
RNPD-01	**Macro RF Nominal Planning & 3D Propagation Modeling**	Macro-level 5G RF network planning using advanced 3D ray-tracing propagation modeling engines, calibrated against high-resolution 5-meter clutter models and Digital Terrain Models (DTM). Sizing and generation of nominal candidate search rings across high-density suburban and rural barangay centroids; multi-carrier capacity dimensioning for Band n50 (60 MHz NCR / up to 80 MHz Provincial, Software-Configurable 40/50/60/80 MHz TDD); coverage prediction mapping (RSRP, SINR, Best Server, Downlink/Uplink Throughput); and inter-site distance (ISD) optimization. • **Inclusions:** 3D ray-tracing simulation, clutter model calibration, nominal site planning, search ring generation. • **Exclusions:** Procurement of specialized external digital terrain data licenses (provided by Operator).	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon formal Operator approval of the Nominal RF Network Design Package.	**Nominal RF Network Design Package & Ray-Tracing Simulation Dossier**
RNPD-02	**Cell Design Data (CDD) Engineering & Parameter Allocation**	Engineering, generation, and validation of detailed site-specific Cell Design Data (CDD) sheets: cell naming taxonomy, gNodeB IDs, Physical Cell ID (PCI) planning adhering to modulo-3 and modulo-30 collision rules, Root Sequence Index (RSI), Tracking Area Codes (TAC), Neighbor Cell Lists (NCL / ANR), PRACH configuration, absolute radio-	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon validation and successful upload of CDD parameters into the network EMS/OSS.	**Approved Cell Design Data (CDD) Sheet & Parameter Script**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		frequency channel numbers (ARFCN), per-carrier transmit power, and antenna configuration parameters (azimuth, mechanical tilt, eRET ranges). • **Inclusions:** CDD parameter engineering, PCI planning, neighbor list generation, configuration script generation. • **Exclusions:** Core network routing configuration (handled under Lot 3).							
CAT-04	**DOMAIN: TELECOM LOGISTICS & MATERIAL HANDLING**								
LOG-01	**Central & Regional Warehousing, Buffer Staging & Serialized Asset Tagging**	Receipt of OEM shipments at Port of Manila / Batangas; bonded climate-controlled warehousing in central Mega Manila and regional hubs (North Luzon, South Luzon, Visayas, Mindanao); unboxing, physical inspection, and QA damage checking; barcode asset tagging at serial-number level; updating Operator ERP/WMS inventory records; and maintaining segregated buffer stock. • **Inclusions:** Warehouse facility, security, ERP tracking, barcode tagging, climate control, buffer stock management. • **Exclusions:** Port customs duties and VAT taxes (direct Operator responsibility).	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon material receipt, inspection, and barcode registration into inventory ERP.	**Serialized Inventory Receiving Report & Barcode Registry**
LOG-02	**Site BOM Kitting, Staging & Pre-Assembly Services**	Assembly and kitting of site-specific deployment packages: picking 1x BBU, 3x 8T8R RRUs, 3x macro antennas/AAUs, optical patchcords, DC power cables, Type 2 SPDs, grounding kits, and hardware accessories; pre-cutting cable lengths based on approved TSSR; labeling; staging in protective crates; and performing bench	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon staging and issuance of Site Dispatch Slip.	**Site BOM Kitting Slip & Staging Inspection Checklist**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		test pre-checks and firmware pre-flashing. • **Inclusions:** Picking, staging, pre-cutting cables, labeling, firmware pre-flashing, packaging into site kits. • **Exclusions:** Supply of missing OEM radio equipment (OEM responsibility).							
LOG-03	**Regional Transport & Final-Mile Rough-Road Site Hauling**	Transportation from central warehouse to regional hubs via heavy freight, followed by dedicated final-mile transport to remote barangay site locations utilizing heavy-duty 4x4 trucks, motorized bancas/barges (for island sites), or manual porterage across unpaved rough-road terrain; comprehensive in-transit cargo insurance; and on-site offloading. • **Inclusions:** Long-haul freight, final-mile 4x4 hauling, manual porterage, transit insurance, on-site offloading. • **Exclusions:** Helicopter airlifts (applicable only under extraordinary force majeure call-out).	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon verified on-site delivery and execution of Material Receiving Report (MRR).	**Signed Material Receiving Report (MRR) by on-site supervisor**
CAT-05	**DOMAIN: TELECOM INSTALLATION, COMMISSIONING & INTEGRATION**								
TI-01	**Tower Rigging & Mechanical Antenna / RRU Installation**	Mechanical mounting by certified riggers: hoisting and installing 3x macro high-gain antennas and 3x 8T8R RRUs on tower antenna mounting brackets (AMB) at specified AGL centerlines (30m to 45m); setting precise mechanical downtilt and sector azimuths (0°, 120°, 240°) using calibrated digital inclinometers and compasses; and installing AISG 2.0 eRET control cables. • **Inclusions:** Tower rigging crews, safety equipment, hoisting gear, AMB mounting, antenna and RRU installation. • **Exclusions:** Supply of steel antenna mounting brackets	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	50% upon mechanical mounting completion; 50% upon joint rigging inspection.	**Mechanical Rigging & Antenna Alignment Inspection Sheet**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		(AMB) (governed under Lot 1 hardware).							
TI-02	**RF, Optical Fiber & DC Power Cabling Installation**	Routing, fastening, and dressing armored multi-core optical fiber jumpers and DC power cables along tower cable ladders and horizontal gantries; installing cold-shrink weatherproof boots on all 7/16 DIN or 4.3-10 RF connectors; installing Type 2 DC Surge Protection Devices (SPD) at tower top and base; and bonding 3-point grounding kits to the external ground busbar. • **Inclusions:** Cable routing, cleating, hydraulic crimping, weatherproofing, SPD installation, grounding bond. • **Exclusions:** Internal civil cabling of third-party shelters.	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	50% upon cabling installation; 50% upon weatherproofing and grounding sign-off.	**Cabling Installation & Weatherproofing Quality Checklist**
TI-03	**Baseband Unit (BBU) Installation & Cabinet Wiring**	19" rack installation of 5G Baseband Unit (BBU) in outdoor equipment cabinet or shelter; dual -48V DC power wiring to dedicated distribution breakers; routing backhaul transmission optical patchcords to fiber demarcation router; and installing GNSS/GPS antenna and surge arrester with unobstructed sky view for PTP/SyncE clock synchronization. • **Inclusions:** Rack mounting, DC wiring, optical patchcord routing, GPS antenna installation, labeling. • **Exclusions:** Supply of DC rectifier plant (governed under Lot 2).	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon BBU rack installation and DC power-up.	**BBU Installation & Cabinet Wiring Verification Sheet**
TI-04	**Telecom Commissioning, Sweep Testing & Power-Up**	DC power turn-up, voltage and polarity verification; firmware loading and diagnostic testing; optical transceiver Rx/Tx power level verification; and calibrated RF sweep testing measuring Return Loss (>= 16.5 dB) and VSWR (<= 1.35:1) across all antenna ports and jumpers using a calibrated SiteMaster	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon successful sweep testing and commissioning approval.	**Commissioning Test Log & VSWR/Return Loss Sweep Traces**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		analyzer. • **Inclusions:** Commissioning turn-up, firmware loading, optical power check, VSWR sweep testing, test trace logs. • **Exclusions:** Rectification of OEM internal hardware manufacturing defects (OEM warranty scope).							
TI-05	**gNodeB Network Integration & OSS Commercial On-Air**	Transmission IP integration (VLAN, BGP, IPsec); establishing S1-U / N2/N3 control and user plane interfaces to 5G Core (AMF/UPF); synchronization with TelTech EMS/OSS; activating 3 sectors on Band n50 (60 MHz NCR / up to 80 MHz Provincial, Software-Configurable 40/50/60/80 MHz TDD); clearing all active alarms; and achieving stable commercial on-air operational state. • **Inclusions:** IP configuration, Core integration, OSS synchronization, carrier activation, alarm clearance. • **Exclusions:** Core network UPF dimensioning (governed under Lot 3).	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon site on-air declaration and alarm-free verification.	**Commercial On-Air Certificate & OSS Alarm-Free Screenshot**
CAT-06	**DOMAIN: SINGLE SITE VERIFICATION & ACCEPTANCE**								
ACC-01	**Mechanical, Civil & Electrical Site Acceptance (ATP-A)**	Joint physical on-site inspection: verifying bolt torque marks on tower mounts, cable ladder fastening, cable tray grounding, weatherproofing seal integrity, electrical breaker labeling, 3-point earth ground resistance audit (confirming <= 5.0 Ohms), compound housekeeping, and punchlist compilation and clearance. • **Inclusions:** Joint physical audit, torque inspection, grounding verification, punchlist clearance. • **Exclusions:** Radio performance verification (covered under ATP-B and SSV).	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon joint ATP-A punchlist clearance and sign-off.	**Signed ATP-A Certificate & Punchlist Clearance Sheet**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
ACC-02	**Active Telecom & Functional Acceptance (ATP-B)**	Remote and local active functional testing: verifying 8T8R MU-MIMO operation, electronic downtilt (eRET) remote control functionality via OSS, power fail / mains failure alarm simulation, optical fiber failover test, and carrier shutdown/restart diagnostic tests to ensure robust autonomic operation. • **Inclusions:** Remote EMS diagnostic testing, alarm simulation, eRET control validation, MIMO verification. • **Exclusions:** Physical climbing inspection (covered under ATP-A).	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon joint ATP-B sign-off.	**Signed ATP-B Functional Acceptance Certificate**
ACC-03	**Single Site Verification (SSV) & RF Call Testing**	On-ground field testing using 5G test UE and scanner: stationary throughput tests at near, mid, and cell-edge points per sector (validating >= 450 Mbps peak DL, >= 50 Mbps cell-edge DL, >= 15 Mbps cell-edge UL); sector swap validation; intra-site handover execution; latency measurement (<= 18 ms RTT); and compilation and sign-off of the formal SSV Report. • **Inclusions:** Stationary throughput tests, sector swap check, handover test, ping test, SSV report compilation. • **Exclusions:** Multi-site cluster drive testing (covered under Category 7).	Per Site	1,000	5,000	[Bidder Quote]	[Formula]	100% upon formal SSV Report approval by Operator.	**Approved Single Site Verification (SSV) Report & Log Files**
CAT-07	**DOMAIN: CLUSTER RADIO NETWORK OPTIMIZATION**								
RNO-01	**Pre-Launch Cluster Drive Testing & RF Baseline Audit**	Continuous scanner and test UE drive testing across contiguous clusters of 10–20 sites; capturing RSRP, SINR, CQI, throughput, and coverage overlap; and identifying pilot pollution zones, coverage holes, and excessive inter-cell interference. • **Inclusions:** Drive	Per Site Share (or Per Cluster)	1,000	5,000	[Bidder Quote]	[Formula]	50% upon completion of cluster drive test and baseline report submission.	**Cluster Drive Test Raw Logs & RF Baseline Audit Report**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		test vehicle, scanner, test UEs, GPS logging, drive test raw log processing, baseline report. • **Exclusions:** Physical antenna re-orientation (governed under RNO-02).							
RNO-02	**Cluster Physical & Soft Parameter Optimization**	Iterative optimization across cluster: electronic remote electrical tilt (eRET) adjustments to control cell-edge interference and shape coverage; azimuth and mechanical tilt adjustments where indicated; PCI re-planning to resolve mod-3/mod-30 clashes; neighbor list (NCL) cleanup; handover threshold tuning; and uplink power control adjustments. • **Inclusions:** eRET tuning, PCI re-planning, neighbor tuning, handover optimization, parameter upload script. • **Exclusions:** Major tower antenna re-heightening (governed by call-out rate card).	Per Site Share (or Per Cluster)	1,000	5,000	[Bidder Quote]	[Formula]	50% upon parameter implementation and verification in OSS.	**Cluster Parameter Optimization Record & Delta Script**
RNO-03	**Cluster Acceptance KPI Validation & Final Sign-Off**	Post-tuning verification drive test and OSS 14-day statistical performance audit: validating cluster Radio Availability >= 99.95%, Session Setup Success Rate (CSSR) >= 99.60%, Session Drop Rate (CDDR) <= 0.15%, cluster average DL throughput >= 450 Mbps, cell-edge DL >= 50 Mbps; and submission of formal Cluster Acceptance Dossier. • **Inclusions:** Verification drive test, 14-day OSS KPI statistical audit, benchmark reporting, formal sign-off. • **Exclusions:** Ongoing post-commercial operations optimization (covered under Lot 6 Managed Services).	Per Cluster (65 Clusters Phase 1 / 325 Clusters Program)	65	325	[Bidder Quote]	[Formula]	100% upon formal sign-off of Cluster Acceptance Certificate.	**Approved Cluster Acceptance Certificate & 14-Day KPI Audit Report**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
CAT-08	**DOMAIN: OPERATIONS & MAINTENANCE (O&M) / MANAGED SERVICES**								
MS-NOC-01	**24/7/365 Central NOC Tier-1 Active Surveillance & Alarm Monitoring**	Continuous 24/7/365 active monitoring of all active and passive network elements across RAN, Core, Power, and Transmission; automated alarm correlation and noise suppression; ticket logging within 5 minutes of alarm onset; and issuance of Daily Operational Flash Report at 06:00 PHT. • **Inclusions:** 24/7 Tier-1 NOC staffing, automated correlation, trouble ticket creation, daily flash reporting. • **Exclusions:** Third-party commercial power utility outages outside contractor control (tracked as excusable SLA).	Per Site / Month	1,000	5,000	[Bidder Quote]	[Formula]	Monthly in arrears upon SLA performance certification.	**Daily Flash Reports, Monthly NOC Surveillance Dossier & Availability Report**
MS-NOC-02	**Incident Management, Triage & Automated Trouble Ticketing**	End-to-end incident management: automated ticketing via OSS/ITSM; severity classification (Sev 1 Critical to Sev 4 Warning); technician dispatch initiation within 15 minutes for Sev 1 incidents; bridge coordination; and stakeholder notification. • **Inclusions:** ITSM platform, ticket lifecycle tracking, escalation dispatch, MTTR SLA auditing. • **Exclusions:** Operator customer service call center operations (BSS scope).	Per Site / Month	1,000	5,000	[Bidder Quote]	[Formula]	Monthly in arrears.	**Monthly Trouble Ticket Log, MTTR Audit & Dispatch Statistics**
MS-NOC-03	**Tier-2 Remote Technical Diagnostics, Fault Recovery & Config Management**	24/7 remote engineering support: soft and hard resets, remote diagnostic sweeps, card re-initialization, backup configuration restoration, transmission routing failover, and software rollback to clear network faults without field truck rolls. • **Inclusions:** Tier-2 remote engineers, remote script execution, transmission path failover, config	Per Site / Month	1,000	5,000	[Bidder Quote]	[Formula]	Monthly in arrears.	**Remote Fault Resolution Log & Configuration Change Audit**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		restoration. • **Exclusions:** Physical on-site hardware replacement (covered under FLM).							
MS-NOC-04	**Tier-3 Engineering Escalation & OEM TAC Management**	Advanced engineering escalation for complex system bugs, protocol crashes, and persistent hardware/software anomalies: case logging with OEM TAC (Level 3), trace log collection, bug patch testing, software patch deployment, and Root Cause Analysis (RCA) reporting within 24 hours of major incident restoration. • **Inclusions:** OEM TAC case management, trace capture, patch verification, RCA report compilation. • **Exclusions:** OEM base software annual license royalties (governed under Software AMC).	Per Site / Month	1,000	5,000	[Bidder Quote]	[Formula]	Monthly in arrears.	**Formal Root Cause Analysis (RCA) Reports & OEM TAC Case Log**
MS-FLM-01	**24/7 Field First Line Maintenance (FLM) Emergency Response**	Provision of dedicated, nationwide 24/7 mobile field maintenance crews equipped with 4x4 vehicles, complete safety gear, calibrated test instruments, and emergency restoration tools. Strict compliance with mandated MTTR SLAs: Severity 1 (Critical - Site Outage): Response <= 15 mins, MTTR <= 2.0h (Urban) / <= 4.0h (Rural); Severity 2 (Major - Sector Down / Power Degraded): Response <= 30 mins, MTTR <= 4.0h (Urban) / <= 6.0h (Rural); Severity 3 (Minor - Non-Service Affecting): Response <= 1h, MTTR <= 12.0h (Urban) / <= 24.0h (Rural); Severity 4 (Warning / Planned): Response <= 4h, MTTR <= 48.0h (Urban) / <= 72.0h (Rural). • **Inclusions:** Dedicated 24/7	Per Site / Month	1,000	5,000	[Bidder Quote]	[Formula]	Monthly in arrears, subject to SLA service credit adjustments.	**Monthly FLM Incident Log, Restoration Work Orders & MTTR SLA Report**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		crews, 4x4 maintenance vehicles, tools, fuel, emergency call-outs. • **Exclusions:** Force majeure acts of God (typhoon damage recovery governed under Call-Out Rate Card).							
MS-FLM-02	**Active Telecom Hardware Replacement & Field Swapping**	On-site replacement of damaged or defective active telecom hardware: RRUs, BBUs, baseband processing cards, optical SFP transceivers, RF jumpers, and GPS antennas, utilizing pre-staged regional spare parts float to achieve fast-track MTTR restoration. • **Inclusions:** Technician labor, de-installation, replacement installation, power-up test, return voucher. • **Exclusions:** Cost of out-of-warranty hardware destruction caused by third-party vandalism.	Included in FLM Monthly Retainer / Per Site-Month	1,000	5,000	[Bidder Quote]	[Formula]	Monthly in arrears.	**Hardware Replacement Slip & Defective Unit Return Voucher**
MS-FLM-03	**Passive Power & Infrastructure Emergency Restoration**	Rapid restoration of passive site elements: resetting/replacing tripped DC distribution breakers, rectifying rectifier module faults, clearing battery management system (BMS) lockouts, replacing blown AC/DC surge protection devices (SPDs), and diagnosing automatic transfer switch (ATS) failures. • **Inclusions:** Electrical diagnosis, breaker reset/replacement, SPD swapping, BMS diagnostic clearing. • **Exclusions:** Replacement of entire battery bank destroyed by external flood (insurance scope).	Included in FLM Monthly Retainer / Per Site-Month	1,000	5,000	[Bidder Quote]	[Formula]	Monthly in arrears.	**Passive Infrastructure Restoration Work Order**
MS-FLM-04	**Emergency Fiber Backhaul Splicing & Optical Link Restoration**	Emergency mobilization for fiber transmission outages: OTDR optical fault localization, optical cable drop replacement, mechanical or fusion splicing, and optical splice enclosure re-	Included in FLM Monthly Retainer / Per Site-Month	1,000	5,000	[Bidder Quote]	[Formula]	Monthly in arrears.	**Optical OTDR Trace Report & Splicing Restoration Certificate**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		sealing to restore backhaul transport within mandated MTTR. • **Inclusions:** OTDR fault tracing, fusion splicing labor, splice enclosure resealing, link test. • **Exclusions:** Major fiber spans exceeding 500 meters (governed by Call-Out Rate Card).							
MS-PM-01	**Quarterly Telecom Equipment Preventive Maintenance (Module A)**	Execution of 100% quarterly routine telecom PM: visual inspection of BBU/RRU chassis, optical fiber connector cleaning and inspection, RF jumper weatherproofing boot inspection, cable clamp torque verification, optical SFP power level verification, and dust filter cleaning. • **Inclusions:** Quarterly on-site inspection, connector cleaning, weatherproofing audit, SFP level audit, PM report. • **Exclusions:** Major equipment overhaul.	Per Site / Quarter (or Monthly share)	1,000	5,000	[Bidder Quote]	[Formula]	Quarterly upon submission of 100% PM checklists and mobile audit photos.	**Quarterly Telecom PM Report & Mobile App Timestamped Audit Dossier**
MS-PM-02	**Quarterly DC Power & LiFePO4 Battery PM (Module B)**	Comprehensive quarterly electrical PM: rectifier module load sharing verification, DC output float/equalize voltage calibration, terminal screw torque verification, LiFePO4 battery BMS diagnostic sweep, cell voltage and internal resistance measurement, and 1-hour controlled battery discharge autonomy test under load. • **Inclusions:** Rectifier calibration, BMS diagnostic sweep, internal resistance test, 1-hour load test. • **Exclusions:** Battery module replacement (handled under RMA / Spares float).	Per Site / Quarter	1,000	5,000	[Bidder Quote]	[Formula]	Quarterly upon submission of Battery Health Dossier.	**DC Power & Battery PM Audit Report + Cell Health Telemetry**
MS-PM-03	**Quarterly Earth Grounding & Lightning Protection Audit (Module C)**	Precision ground resistance audit using 3-point fall-of-potential test instrument (confirming earth ground resistance <= 5.0 Ohms); Cadweld connection inspection; copper	Per Site / Quarter	1,000	5,000	[Bidder Quote]	[Formula]	Quarterly upon submission of Grounding Audit Report.	**Earth Ground Resistance Test Certificate with Instrument Calibration Tag**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		busbar bolt torque check; corrosion audit; and inspection of air terminals, down-conductors, and surge arresters. • **Inclusions:** 3-point fall-of-potential test, Cadweld check, busbar torque check, lightning arrester audit. • **Exclusions:** Civil construction of new ground ring (if soil conditions deteriorate; governed by civil rate card).							
MS–PM–04	**Bi-Annual Tower Structure, Rigging & Aviation Light Audit (Module D)**	Specialized climbing inspection conducted twice per year: physical inspection of tower steel members, monopole climbing ladder, fall arrest cable, AMB mounting bracket bolt torque verification, antenna azimuth/tilt physical check, cable ladder clamping check, and inspection of CAAP-compliant aviation obstruction light and photocell controller. • **Inclusions:** Certified climbing riggers, structural member inspection, bolt torque audit, CAAP light audit. • **Exclusions:** Tower painting / rust rehabilitation (governed under civil facility maintenance).	Per Site / Semi-Annual	1,000	5,000	[Bidder Quote]	[Formula]	Semi-annually upon submission of Tower Rigging Audit Report.	**Bi-Annual Tower Climbing & Rigging Audit Report + High-Res Photos**
MS–PM–05	**Quarterly Site Compound, Civil & Environmental Upkeep (Module E)**	Routine civil compound maintenance: inspection and repair of chain-link perimeter fencing and barbed wire; greasing and servicing gate locking mechanisms; weed cutting, vegetation clearance, and chemical weed-killing within a 3-meter perimeter of compound; drainage channel clearing; and pest/rodent proofing of equipment cabinet cable entries. • **Inclusions:** Compound weeding, perimeter clearing, fence repair, lock lubrication, cable entry pest sealing. • **Exclusions:** Major civil perimeter wall re-construction post-landslide.	Per Site / Quarter	1,000	5,000	[Bidder Quote]	[Formula]	Quarterly upon compound inspection sign-off.	**Compound Upkeep Audit Checklist & Before/After Photos**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
MS-SPM-01	**Regional Spare Parts Buffer Float Management (SPMS)**	Operation of climate-controlled, secure regional spare parts depots (Mega Manila, North Luzon, South Luzon, Visayas, Mindanao); managing and maintaining an agreed 2.0% active hardware spares float (BBUs, RRUs, SFPs, rectifiers, batteries, cables); tracking serial numbers via WMS; and conducting monthly physical cycle counts. • **Inclusions:** Regional storage depots, inventory management, serialized tracking, monthly cycle counts. • **Exclusions:** Procurement of additional replacement spares beyond contractual float (Operator capex).	Per Site / Month	1,000	5,000	[Bidder Quote]	[Formula]	Monthly in arrears.	**Monthly Spare Parts Float Status Report & Cycle Count Audit**
MS-SPM-02	**Advance Hardware Replacement (AHR) & Expedited Field Delivery**	Rapid logistics dispatch delivering replacement hardware to field maintenance crews: Next-Business-Day (NBD) standard delivery nationwide, or expedited 4-hour emergency delivery for major macro hubs and critical transmission nodes. • **Inclusions:** Expedited dispatch, emergency courier, chain of custody documentation, field delivery receipt. • **Exclusions:** Standard scheduled routine replenishment.	Included in SPM Retainer / Per Site-Month	1,000	5,000	[Bidder Quote]	[Formula]	Monthly in arrears.	**Hardware Dispatch Manifest & Proof of On-Site Delivery**
MS-SPM-03	**Return Merchandise Authorization (RMA) & OEM Factory Repair Logistics**	Managing the reverse logistics lifecycle for defective hardware: retrieving faulty units from field crews, conducting triage, generating RMA requests, managing packaging and export/import customs for OEM repair/replacement, tracking turnaround time (TAT <= 30 calendar days), and returning repaired units to the regional buffer float. • **Inclusions:** Defective unit triage, RMA processing, shipping to OEM	Included in SPM Retainer / Per Site-Month	1,000	5,000	[Bidder Quote]	[Formula]	Monthly in arrears.	**RMA Lifecycle Tracking Dossier & OEM Repair Turnaround Log**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		repair facility, tracking, customs clearance. • **Exclusions:** Out-of-warranty OEM repair billing charges (passed through at actual OEM repair cost).							
MS–RNO–01	**Continuous Radio Network Optimization & Weekly KPI Surveillance**	Ongoing network-wide RF quality surveillance: daily and weekly statistical KPI tracking (Radio Availability, CSSR, CDDR, average throughput, PRB utilization, cell-edge speeds); identification and root-cause analysis of Worst Performing Cells (WPC); dynamic parameter fine-tuning; and weekly optimization meetings. • **Inclusions:** Dedicated RF optimization engineers, weekly WPC triage, parameter tuning scripts, KPI governance. • **Exclusions:** Major network replanning requiring addition of new cell sites (expansion capex).	Per Site / Month	1,000	5,000	[Bidder Quote]	[Formula]	Monthly in arrears.	**Weekly Worst Performing Cell (WPC) Report & Monthly Network KPI Dossier**
MS–RNO–02	**Interference Hunting, PIM Mitigation & Customer Complaint Field Audit**	Field RF investigation for degraded sectors: spectrum analyzer sweeps to detect and locate external interference sources (jamming, unauthorized boosters, microwave leakage); Passive Intermodulation (PIM) testing and remediation on antenna lines; and on-site drive validation for VIP or high-churn barangay clusters. • **Inclusions:** Spectrum analyzer field sweeps, PIM testing, interference source triangulation, customer complaint verification. • **Exclusions:** Legal action against illegal booster operators (handled by NTC / Operator Legal).	Per Site / Month	1,000	5,000	[Bidder Quote]	[Formula]	Monthly in arrears.	**Interference Investigation Dossier & Spectrum Sweep Traces**
MS–FAC–01	**Passive Tower & Civil Facility Maintenance**	Structural preservation of passive site assets: rust scraping, cold galvanizing compound	Per Site / Month	1,000	5,000	[Bidder Quote]	[Formula]	Monthly in arrears.	**Monthly Facility Maintenance Log & Civil Condition Report**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		application, and repainting of steel members in corrosive/coastal environments; repairing cracked equipment plinths; re-caulking cable entry boots; and maintaining access road drainage and clear-way. • **Inclusions:** Anti-corrosion touch-up, cold galvanizing, plinth crack sealing, cable entry maintenance. • **Exclusions:** Full tower structural re-erection post-catastrophic failure.							
MS-PWR-01	**Energy, Utility Grid (Meralco/Coop) Coordination & Genset PM**	Power management coordination: monitoring utility power reliability, validating commercial meter readings, coordinating with Meralco/electric cooperatives for transformer fuse restoration, and performing routine 250-hour preventive servicing on backup diesel generator sets (engine oil, oil filter, fuel filter, air cleaner replacement, coolant check). • **Inclusions:** Utility outage coordination, meter reading audit, 250-hr generator PM servicing, coolant/filter checks. • **Exclusions:** Commercial grid electric utility consumption charges (Operator pays utility bills).	Per Site / Month	1,000	5,000	[Bidder Quote]	[Formula]	Monthly in arrears.	**Monthly Energy Consumption Audit & Genset PM Service Log**
MS-SW-01	**Software Support, Firmware Patching & Cybersecurity Hardening**	Scheduled execution of software maintenance: applying OEM-approved firmware updates and feature patches during maintenance windows (00:00–05:00); implementing CVE security patches and hardening configs; executing weekly automated BBU and Core configuration backups; and maintaining offsite configuration archives. • **Inclusions:** Firmware deployment, CVE	Per Site / Month	1,000	5,000	[Bidder Quote]	[Formula]	Monthly in arrears.	**Monthly Patch Deployment Log & Configuration Backup Verification Certificate**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		security updates, maintenance window execution, automated backups. • **Exclusions:** New major software feature releases requiring feature license fee purchase (Lot 1/Lot 3 scope).							
CAT-09	**DOMAIN: AD-HOC CALL-OUT SERVICES RATE CARD**								
CALL-01	**Specialized Tower Rigging Call-Out Service**	Ad-hoc mobilization of a certified tower climbing crew to execute non-standard physical antenna modifications: adjusting mechanical tilt, azimuth, or antenna mounting height upon structural reconfiguration or post-typhoon realignment. • **Inclusions:** Rigging crew, climbing tackle, digital compass/inclinometer alignment, before/after photos. • **Exclusions:** Supply of new replacement antennas.	Per Sector / Per Man-Day	100	500	[Bidder Quote]	[Formula]	100% upon completion and photographic verification.	**Rigging Work Order Completion Sheet & Re-alignment Certificate**
CALL-02	**Emergency Typhoon Disaster Recovery Crew**	Rapid deployment of a specialized 4-man disaster recovery crew equipped with heavy-duty 4x4 vehicle, chainsaws, winches, heavy rigging tackle, and portable generator to clear fallen trees, re-erect toppled poles, and execute emergency structural stabilization post-typhoon. • **Inclusions:** 4-man emergency crew, 4x4 vehicle, chainsaw, winch, generator, debris clearing gear. • **Exclusions:** Supply of complete replacement steel towers.	Per Crew / Per Day	50	250	[Bidder Quote]	[Formula]	100% upon daily crew timesheet approval by Operator.	**Disaster Recovery Timesheet & Field Activity Report**
CALL-03	**Mobile Emergency Generator Deployment & Continuous Run**	Delivery, positioning, cabling, and commissioning of a mobile trailer-mounted 15–20 kVA diesel generator at a damaged site during extended utility grid blackouts; includes 48 hours of continuous operation, continuous refueling, and operator surveillance.	Per Deployment Event (48-hr block)	50	250	[Bidder Quote]	[Formula]	100% upon completion of emergency run.	**Emergency Power Log & Generator Run-Hour Telemetry**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		• **Inclusions:** Trailer genset delivery, connection cables, 48 hours continuous running, fuel, technician operator. • **Exclusions:** Operation beyond 48 hours (billed as consecutive additional 48-hr call-out blocks).							
CALL-04	**Long-Span Optical Fiber Cable Restoration & Splicing**	Emergency or planned restoration of major backhaul fiber cable cuts exceeding 500 meters: stringing replacement 24-core armored aerial or underground fiber cable, installing suspension clamps, fusion splicing at both splice closures, and testing optical loss. • **Inclusions:** Cable stringing labor, suspension clamps, 2x fusion splice enclosures, OTDR testing. • **Exclusions:** Supply of raw fiber optic cable (provided from operator inventory or billed separately).	Per Meter / Per Span	10,000	50,000	[Bidder Quote]	[Formula]	100% upon OTDR bi-directional loss sign-off.	**Bi-Directional OTDR Loss Trace & Splicing Certificate**
CALL-05	**Drone Structural, Panoramic & Thermal Imaging Survey**	High-resolution aerial inspection utilizing drone equipped with 4K optical zoom and radiometric thermal sensor: auditing tower structural integrity, inspecting hard-to-reach antenna brackets, and detecting hotspot anomalies on power joints and cables. • **Inclusions:** Licensed drone pilot, 4K video, high-res photos, radiometric thermal imaging, analysis report. • **Exclusions:** Physical repairs to identified anomalies.	Per Site Flight	50	250	[Bidder Quote]	[Formula]	100% upon submission of 4K video, photos, and thermal report.	**High-Resolution Aerial Video/Photo Suite & Thermal Analysis Report**
CALL-06	**Tower Structural Strengthening & Leg Reinforcement**	Physical structural modification of existing tower or monopole: installing steel reinforcement members, flange stiffeners, cross-bracing, or guy wire tensioning based on approved PE structural engineering calculations. • **Inclusions:** Steel fabrication, rigging	Per Tower / Lump Sum based on DED	20	100	[Bidder Quote]	[Formula]	50% upon material delivery; 50% upon PE structural sign-off.	**As-Built Structural Strengthening Drawings & PE Endorsement**

Item Ref	Category / Title	Detailed Technical Scope & Boundaries	UoM	Phase 1 Qty	Program Qty	Unit Price (USD)	Unit Price (PHP @ 55)	Billing Milestone Trigger	Deliverable Artifact
		installation, bolt torquing, cold galvanizing, PE sign-off. • **Exclusions:** Supply of primary raw steel members (billed based on approved DED BoQ).							
CALL-07	**Specialized PIM Hunting & Multi-Sector Interference Remediation**	Advanced on-site PIM testing utilizing specialized 2x20W PIM analyzer: identifying internal and external PIM sources, replacing corroded RF jumpers/connectors, and applying RF shielding tape/materials to achieve PIM <= -155 dBc. • **Inclusions:** Calibrated 2x20W PIM test gear, RF sweep, connector re-termination, shielding materials, sweep report. • **Exclusions:** Replacement of damaged antenna internal phase shifters (OEM warranty).	Per Site / Day	30	150	[Bidder Quote]	[Formula]	100% upon achieving compliant PIM sweep trace.	**Calibrated PIM Sweep Measurement Report**
CALL-08	**Ad-Hoc Emergency Fuel Delivery & Mobile Refueling**	Rapid procurement and transport of clean Euro-4 automotive diesel fuel in certified fuel containers to remote sites during prolonged grid emergencies, including manual transfer to day tank. • **Inclusions:** Fuel procurement, dedicated transport, manual pumping into tank, fuel quality test. • **Exclusions:** Cost of bulk commercial diesel fuel (reimbursed at actual pump receipt price).	Per Liter Delivered / Per Trip	20,000	100,000	[Bidder Quote]	[Formula]	100% upon fuel receipt voucher sign-off.	**Fuel Purchase Receipt & Fuel Delivery Voucher**

5. SERVICE LEVEL AGREEMENTS (SLAS) & LIQUIDATED REMEDIES

Domain	Metric Identifier	Mandatory SLA Target	Minimum Operating Floor	Service Credit / Liquidated Remedy
Rollout Delivery	Turnkey Milestone Schedule	**1,000 sites in 365 Days**	Milestone baseline	0.5% milestone contract value per week of delay (cap 10%)
Radio Availability	KPI-RAN-01 Network Uptime	**≥ 99.95% Availability**	99.70% Minimum	1.0% monthly O&M fee deduction per 0.1% shortfall
Call Accessibility	KPI-RAN-02 5G Session CSSR	**≥ 99.60% Success**	99.00% Minimum	0.5% monthly O&M fee deduction per 0.2% shortfall
Call Retainability	KPI-RAN-03 Session Drop CDDR	**≤ 0.15% Drops**	0.50% Maximum	0.5% monthly O&M fee deduction per 0.1% excess

Domain	Metric Identifier	Mandatory SLA Target	Minimum Operating Floor	Service Credit / Liquidated Remedy
Cluster Throughput	KPI-RAN-04 Loaded Sector DL	**≥ 450 Mbps Average**	380 Mbps Floor	Mandatory RF re-tuning & re-drive within 5 days
Cell-Edge Speed	KPI-RAN-05 Cell-Edge DL Speed	**≥ 50 Mbps DL (95% pts)**	25 Mbps Floor	Mandatory eRET electrical tilt re-optimization
FLM Restoration	MTTR Severity 1 (Urban)	**≤ 2.0 Hours MTTR**	2.0 Hours	0.5% monthly site fee per hour delay beyond MTTR
FLM Restoration	MTTR Severity 1 (Rural)	**≤ 4.0 Hours MTTR**	4.0 Hours	0.25% monthly site fee per hour delay beyond MTTR
Preventive Maint.	Quarterly PM Completion	**100% Completed p.a.**	98.0% Floor	Uncompleted site PMs debited 200% of line item rate
Spare Parts Logistics	Advance Replacement (AHR)	**Next Business Day (NBD)**	NBD Standard	$250 USD penalty per calendar day delay in AHR delivery

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